

ODE Coefficient Synthesis

The coefficient lattice and its staircase

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Abstract

Dimensional constraints generate the Laurent-monomial table $A_{r,k}$ for $(a,b,c) = (L,R,1/C)$. Minimizing the sum of absolute exponents selects its staircase and gives coefficients at every derivative order. Substituting $D = i\omega$ yields the corresponding phasor factors and finite sums.

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Latest on GitHub: <https://github.com/hobnilre/physics-ode-coefficient-synthesis>

1 Deriving the coefficient family

Use the template $aD^2y + bDy + cy = f$ from *The ODE Template and Its Domain Equivalents*, with $D = d/dt$ and constant $a,b,c > 0$. Let T be the time dimension and $U = [c]$, taken as independent dimensions. Equality of term dimensions gives

$$[a] = UT^2, \quad [b] = UT, \quad [c] = U.$$

A coefficient multiplying $D^k y$ must therefore have dimension UT^k . Seek it as a monomial $a^p b^q c^r$. Its dimensions are $U^{p+q+r} T^{2p+q}$, so the exponents must satisfy

$$p + q + r = 1, \quad 2p + q = k.$$

Solving for p and q gives, up to a dimensionless multiplier,

$$A_{r,k}(a,b,c) = a^{k-1+r} b^{2-k-2r} c^r. \quad (1)$$

We take $r,k \in \mathbb{Z}$, so these are Laurent monomials: integer powers may be positive, zero or negative. Dimension matching also permits real r ; integrality is the chosen algebraic class.

For $(a,b,c) = (L,R,1/C)$, equation (1) becomes

$$A_{r,k} = L^{k-1+r} R^{2-k-2r} C^{-r}. \quad (2)$$

For example, at $k = 3$ the choices $r = 0, -1, -2$ give

$$\frac{L^2}{R}, \quad LRC, \quad R^3 C^2.$$

The two dimensional constraints leave one exponent free. They determine a family of coefficients for each order; a selection rule chooses a member.

2 The coefficient table

Table 1 displays equation (2) in two panels. Boxed entries mark the staircase derived in the next section. The ODE uses $k \geq 0$; negative columns extend the dimension rule to repeated integrals, whose time factors are T^{-k} .

$r \backslash k$	-4	-3	-2	-1	0
3	$\frac{1}{L^2 C^3}$	$\frac{1}{LRC^3}$	$\frac{1}{R^2 C^3}$	$\frac{L}{R^3 C^3}$	$\frac{L^2}{R^4 C^3}$
2	$\frac{R^2}{L^3 C^2}$	$\frac{R}{L^2 C^2}$	$\frac{1}{LC^2}$	$\frac{1}{RC^2}$	$\frac{L}{R^2 C^2}$
1	$\frac{R^4}{L^4 C}$	$\frac{R^3}{L^3 C}$	$\frac{R^2}{L^2 C}$	$\frac{R}{LC}$	$\frac{1}{C}$
0	$\frac{R^6}{L^5}$	$\frac{R^5}{L^4}$	$\frac{R^4}{L^3}$	$\frac{R^3}{L^2}$	$\frac{R^2}{L}$
-1	$\frac{R^8 C}{L^6}$	$\frac{R^7 C}{L^5}$	$\frac{R^6 C}{L^4}$	$\frac{R^5 C}{L^3}$	$\frac{R^4 C}{L^2}$
-2	$\frac{R^{10} C^2}{L^7}$	$\frac{R^9 C^2}{L^6}$	$\frac{R^8 C^2}{L^5}$	$\frac{R^7 C^2}{L^4}$	$\frac{R^6 C^2}{L^3}$
-3	$\frac{R^{12} C^3}{L^8}$	$\frac{R^{11} C^3}{L^7}$	$\frac{R^{10} C^3}{L^6}$	$\frac{R^9 C^3}{L^5}$	$\frac{R^8 C^3}{L^4}$

$r \backslash k$	1	2	3	4	5	6
3	$\frac{L^3}{R^5 C^3}$	$\frac{L^4}{R^6 C^3}$	$\frac{L^5}{R^7 C^3}$	$\frac{L^6}{R^8 C^3}$	$\frac{L^7}{R^9 C^3}$	$\frac{L^8}{R^{10} C^3}$
2	$\frac{L^2}{R^3 C^2}$	$\frac{L^3}{R^4 C^2}$	$\frac{L^4}{R^5 C^2}$	$\frac{L^5}{R^6 C^2}$	$\frac{L^6}{R^7 C^2}$	$\frac{L^7}{R^8 C^2}$
1	$\frac{L}{RC}$	$\frac{L^2}{R^2 C}$	$\frac{L^3}{R^3 C}$	$\frac{L^4}{R^4 C}$	$\frac{L^5}{R^5 C}$	$\frac{L^6}{R^6 C}$
0	R	L	$\frac{L^2}{R}$	$\frac{L^3}{R^2}$	$\frac{L^4}{R^3}$	$\frac{L^5}{R^4}$
-1	$\frac{R^3 C}{L}$	$R^2 C$	LRC	$L^2 C$	$\frac{L^3 C}{R}$	$\frac{L^4 C}{R^2}$
-2	$\frac{R^5 C^2}{L^2}$	$\frac{R^4 C^2}{L}$	$R^3 C^2$	$LR^2 C^2$	$L^2 RC^2$	$L^3 C^2$
-3	$\frac{R^7 C^3}{L^3}$	$\frac{R^6 C^3}{L^2}$	$\frac{R^5 C^3}{L}$	$R^4 C^3$	$LR^3 C^3$	$L^2 R^2 C^3$

Table 1. The coefficient table for $(a, b, c) = (L, R, 1/C)$, with $r = 3, \dots, -3$ and $k = -4, \dots, 6$. Blue boxes select $r_k = \lfloor (2 - k)/2 \rfloor$.

The whole table follows from two ratios:

$$A_{r,k+1} = \frac{L}{R} A_{r,k}, \quad A_{r-1,k} = \frac{R^2 C}{L} A_{r,k}. \quad (3)$$

Thus one step right contributes a time factor, while one step down contributes a dimensionless factor.

3 Selecting the stairs

Choose the integer r that minimizes the sum of absolute exponents in (L, R, C) :

$$F_k(r) = |k - 1 + r| + |2 - k - 2r| + |r|.$$

If two choices tie, choose the one with a nonnegative exponent of R . The result is

$$r_k = \left\lfloor \frac{2 - k}{2} \right\rfloor = 1 - \left\lceil \frac{k}{2} \right\rceil. \quad (4)$$

To see why, the triangle inequality gives

$$F_k(r) \geq |k - 1| + |2 - k - 2r|.$$

The last term is at least 0 for even k and 1 for odd k . The choice (4) attains this lower bound and makes the exponent of R equal to 0 or 1, respectively. It therefore satisfies both the minimum and the tie rule. This simplicity criterion refers to the declared parameter basis.

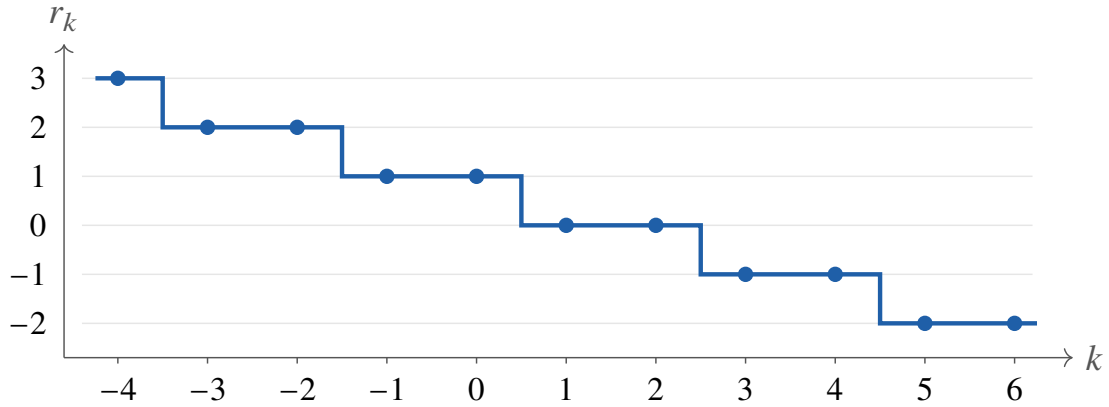


Figure 1. The stairs $r = r_k$. Each horizontal pair uses one value of r ; the next pair is one row lower. The boxed entries in Table 1 follow this path.

Set $A_k = A_{r_k, k}$. Substituting $k = 2n$ and $k = 2n + 1$ into (2) yields

$$A_{2n} = L^n C^{n-1}, \quad A_{2n+1} = R(LC)^n, \quad n \in \mathbb{Z}. \quad (5)$$

Consequently,

$$A_{k+2} = LC A_k, \quad A_0 = \frac{1}{C}, \quad A_1 = R. \quad (6)$$

Every second coefficient is obtained by multiplying by LC . The even and odd orders are two interleaved geometric sequences.

4 Assembling an ODE

For an integer $N \geq 1$, the staircase coefficients define

$$P_N(D)y = 0, \quad P_N(D) = \sum_{k=0}^N A_k D^k. \quad (7)$$

The first seven coefficients, in ascending derivative order, are

$$(A_0, \dots, A_6) = \left(\frac{1}{C}, R, L, LRC, L^2 C, L^2 RC^2, L^3 C^2 \right).$$

In the original reference symbols the same rule is

$$A_{2n} = a^n c^{1-n}, \quad A_{2n+1} = b(a/c)^n, \quad A_{k+2} = \frac{a}{c} A_k.$$

In particular, the coefficient of D^3 is $A_3 = ab/c$, and direct multiplication shows

$$\frac{ab}{c} D^3 + aD^2 + bD + c = (bD + c) \left(1 + \frac{a}{c} D^2 \right). \quad (8)$$

More generally, pairing adjacent orders gives, for $m \geq 0$,

$$P_{2m+1}(D) = (c + bD) \sum_{j=0}^m \left(\frac{a}{c} D^2 \right)^j. \quad (9)$$

For even order $2m$ with $m \geq 1$, add $a^m c^{1-m} D^{2m}$ to $P_{2m-1}(D)$. These are finite polynomial identities; no convergence assumption is needed.

4.1 Choosing other coefficients

To retain the full family, introduce

$$S = \frac{b^2}{a}, \quad \tau = \frac{a}{b}, \quad \rho = \frac{b^2}{ac}.$$

Then $[S] = U$, $[\tau] = T$ and $[\rho] = 1$, and

$$A_{r,k} = S \tau^k \rho^{-r}, \quad B_k = \sum_{r \in \mathcal{R}_k} w_{r,k} A_{r,k} = S \tau^k \sum_{r \in \mathcal{R}_k} w_{r,k} \rho^{-r}, \quad (10)$$

where $\mathcal{R}_k \subset \mathbb{Z}$ is finite and $w_{r,k}$ are dimensionless constants. Using B_k in $\sum_{k=0}^N B_k D^k y = 0$ gives a finite synthesis from the table; its order is N when $B_N \neq 0$. The staircase is the choice $\mathcal{R}_k = \{r_k\}$ and $w_{r_k,k} = 1$.

Dimension matching constructs the table. Minimizing integer exponents selects the stairs, and the stairs give a coefficient at every order through one recurrence. Finite weighted sums retain the other choices.

5 Using phasors

With the convention $\text{Re}(\hat{y}e^{i\omega t})$, $\omega > 0$, from *The ODE Template and Its Domain Equivalents*, replace D by $i\omega$. For real coefficients B_k and $Q(s) = \sum_{k=0}^N B_k s^k$, $Q(D)y = f$ becomes

$$Q(i\omega)\hat{y} = \hat{f}, \quad Q(i\omega) = \sum_{k=0}^N B_k (i\omega)^k. \quad (11)$$

If $Q(i\omega) \neq 0$, then $\hat{y} = \hat{f}/Q(i\omega)$. For the homogeneous equation, a nonzero phasor requires $Q(i\omega) = 0$.

5.1 The table and stairs in phasor form

Multiply column k of Table 1 by $(i\omega)^k$. Each entry contributes

$$\Phi_{r,k}(\omega) = A_{r,k}(i\omega)^k = S\rho^{-r}(i\omega\tau)^k. \quad (12)$$

The factors i^k cycle through 1, i , -1 , $-i$. Negative columns use the same rule for the harmonic antiderivative, with integration constants specified separately. Weighted sums of these entries give $Q(i\omega)$.

For the staircase, write $\Phi_k = A_k(i\omega)^k$. Equation (5) gives

$$\Phi_{2n} = \frac{1}{C}(-\omega^2 LC)^n, \quad \Phi_{2n+1} = i\omega R(-\omega^2 LC)^n, \quad n \in \mathbb{Z}. \quad (13)$$

Hence $\Phi_{k+2} = -\omega^2 LC \Phi_k$: every two orders reverse the sign and multiply the magnitude by $\omega^2 LC$. Pairing terms as in (9) yields

$$P_{2m+1}(i\omega) = \left(\frac{1}{C} + i\omega R\right) \sum_{n=0}^m (-\omega^2 LC)^n, \quad m \geq 0. \quad (14)$$

For example, $P_3(i\omega) = (1/C + i\omega R)(1 - \omega^2 LC)$. The sum is finite, so no restriction $\omega^2 LC < 1$ is needed.

5.2 Expressing the equation using a derivative

If $u = Dy$, then $\hat{u} = i\omega\hat{y}$. The multiplier relating $\hat{u}$ to $\hat{f}$ is therefore

$$H(\omega) = \frac{Q(i\omega)}{i\omega} = \sum_{k=0}^N B_k (i\omega)^{k-1}, \quad \hat{f} = H(\omega)\hat{u}. \quad (15)$$

The exponent is k for y and $k - 1$ for Dy . Terminal ratios of this form are combined in *Interconnecting Series and Parallel ODEs*, Section 3. The signed-work calculation for the selected $A_3 = LRC$ is developed in *Energy Ledgers for Forced Harmonic ODEs*, Parts 2 and 3.

References

1. H. Nilre and B. C. Herlin (2026). *The ODE Template and Its Domain Equivalents*.
2. H. Nilre and B. C. Herlin (2026). *Interconnecting Series and Parallel ODEs*.
3. H. Nilre and B. C. Herlin (2026). *Energy Ledgers for Forced Harmonic ODEs*.